Past Question - 2079

Section 'A'

Qn:1. How do you plot the function on graph? Determine whether the function $f(x) = x^2$ is injective, surjective or bijective. Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 6$.

→ Solution:

Given, $a_n = 6a_{n-1} - 9a_{n-2}$

By comparing, we get, $c_1 = 6$ and $c_2 = -9$.

The characteristics eq^n is:

$$r^2 - c_1 r - c_2 = 0$$

$$\Rightarrow r^2 - 6r + 9 = 0$$

$$\Rightarrow r^2 - (3+3)r + 9 = 0$$

$$\Rightarrow r^2 - 3r - 3r + 9 = 0$$

$$\Rightarrow r(r-3) - 3(r-3) = 0$$

$$\Rightarrow (r-3)(r-3) = 0$$

$$\therefore r = 3, 3$$

Roots are same i.e. $r_0 = 3$.

By question:

$a_0 = 1$

$\Rightarrow a_0 = \alpha_1 r_0^n + \alpha_2 n r_0^n$

$\Rightarrow 1 = \alpha_1 \cdot 3^0 + \alpha_2 \cdot 0 \cdot 3^0$

$\Rightarrow \alpha_1 = 1$

$a_1 = 6$

$\Rightarrow a_1 = r_0^n \alpha_1 + r_0^n \alpha_2 \cdot n$

$\Rightarrow 6 = \alpha_1 3^1 + \alpha_2 \cdot 1 \cdot 3^1$

$\Rightarrow 6 = 3 + 3\alpha_2$

$\Rightarrow \frac{6-3}{3} = \alpha_2$

$\Rightarrow 1 = \alpha_2$

The general solution of recurrence relation is:

$$a_n = \alpha_1 r_0^n + \alpha_2 n \cdot r_0^n$$

$$\therefore a_n = 3^n + n \cdot 3^n \Rightarrow 3^n(1+n) //$$

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Given, $f(x) = x^2$

for one-to-one:

let $x_1, x_2 \in R$. Then,

$$f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2$$
$$\Rightarrow x_1 = x_2 \text{ or, } x_1 \neq -x_2.$$

which means. if $x_1 = 2$ and $x_2 = -2$ then,

$$f(2) = 2^2 = 4$$
$$f(-2) = -2^2 = 4$$

Here, $f(2) = f(-2)$ but $2 \neq -2$.

Here, different inputs (2 and -2) yeild the same output,
So $f(x)$ is not one-to one.

for onto,

let $f(x) = y = x^2$

or, $x = \sqrt{y} \in R$ for all $y \in R$.

and $f(\sqrt{y}) = (\sqrt{y})^2 = y$.

$\therefore$ y is the image of $\sqrt{y}$.

$\therefore$ f is onto.

But seeing from codomain.

① If codomain is all real numbers. ie. Range of $F(x) \Rightarrow [0, \infty)$ then, it will not produce negative values resulting that negative numbers don't have its mapping. so it is not onto.

② If codomain is $[0, \infty)$ and Domain: R

It will be onto because every non-negative number y has a corresponding x in the domain //

jective: The function is not bijective.

Qn.2 A group of 8 scientist is composed of 5 chemist and 3 biologist. In how many ways a comittee of 5 can be formed that has 3 chemist & 2 biologist? Using mathematical induction prove that

$$1^3+2^3+3^3+\dots+n^3 = \frac{n^2(n+1)^2}{4} \quad \text{for } n \geq 1.$$

→ The number of ways to choose 3 chemists out of 5 can be calculated using combination formula:

$$\binom{n}{k} \text{ or } {}^nC_k = \frac{n!}{k!(n-k)!}$$

where $\binom{n}{k}$ is the no. of ways to choose k elements from n elements.

First calculating no. of ways to choose 3 chemists out of 5.

$$\binom{5}{3} = \frac{5!}{3!(5-3)!} = 10$$

Now, calculating no. of ways to choose 2 biologists out of 3

$$\binom{3}{2} = \frac{3!}{2!\,1!} = 3$$

Now,

no. of ways to form a comittee of 5 scientists that include 3 chemists & 2 biologists is: $10 \times 3 = 30$ //

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→ Let $P(n)$ be the given statement then

$$P(n) = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4} \text{ for } n \geq 1$$

Base case: $P(1) = \frac{1^2(1+1)^2}{4}$

$$= \frac{1 \times 4}{4}$$

$= 1$ which is cube of 1

$\therefore P(1)$ is true

Inductive Step:

Assume the statement is true for some integer k.

$$P(k) = 1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4} \text{ for } k \geq 1 \quad \text{--- (1)}$$

Now, we need to show that the statement is true for $k+1$.

$$P(k+1) = 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4} \quad \text{--- (2)}$$

$$L.H.S = 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3$$

$$= \frac{k^2(k+1)^2}{4} + (k+1)^3$$

$$= \frac{k^2(k+1)^2 + 4(k+1)^3}{4}$$

$$= \frac{(k+1)^2(k^2 + 4(k+1))}{4}$$

$$= \frac{(k+1)^2(k^2 + 4k + 4)}{4}$$

$$= \frac{(k+1)^2(k^2 + 2k + 2k + 4)}{4}$$

$$= \frac{(k+1)^2(k(k+2) + 2(k+2))}{4}$$

$$= \frac{(k+1)^2(k+2)(k+2)}{4}$$

$$= \frac{(k+1)^2(k+2)^2}{4}$$

$= R.H.S$

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Past Question-2080.

Section A.

1. Explain strong induction. What is recursively defined function? Use mathematical induction to prove $7^{n+2}+8^{2n+1}$ is divisible by 5.

→ Strong Induction is a proof technique in mathematics used to show that a statement is true for all natural numbers. In strong induction; we have to prove that the statement is true for the first natural no. (in base case) and we have to assume the statement is true for some natural number k. Then prove it is true for k+1.

A recurively defined function is a function that is defined in terms of itself. This means that to find the value of the function at some point, you need to use the values of the function at smaller points.

Given, $7^{n+2}+8^{2n+1}$ is divisible by 57

Base Case: Let's verify the statement is true for $n=1$.

$$P(1) = 7^{1+2}+8^{2\times1+1}$$
$$= 7^3+8^3$$
$$= 343+512 = 855$$

$855 \% 57 = 15$ which is divisible by 57 so, base case holds true.

Inductive Case: Assume the statement is true for some k, that is, assume:

$P(k) = 7^{k+1}+8^{2k+1}$ divisible by 57

Now let's assume k+1 as the statement. then,

$$P(k+1) = 7^{(k+1)+2}+8^{2(k+1)+1}$$
$$= 7^{k+3}+8^{2k+2+1}$$
$$= 7^{k+3}+8^{2k+3}$$

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$= 7^{k+2+1} + 8^{2k+2+1}$

$= 7^1 \cdot 7^{k+2} + 8^{2k+1} \cdot 8^2$

$= 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1}$

$= 7 \cdot 7^{k+2} + (57+7) \cdot 8^{2k+1}$

$= 7 \cdot 7^{k+2} + 57 \cdot 8^{2k+1} + 7 \cdot 8^{2k+1}$

$= 7 \cdot (7^{k+2} + 8^{2k+1}) + 57 \cdot 8^{2k+1}$

Hence, $7^{k+2} + 8^{2k+1}$ is already divisible which is $P(k)$ statement &
$57 \cdot 8^{2k+1}$ is also divisible by 57 since it is a multiple of 57

$\therefore$ $7^{n+2} + 8^{2n+1}$ is divisible by 57 proved //

State pigeonhole principle. Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 3$.

Pigeonhole principle states that if you have more pigeons than pigeonholes, then atleast one pigeonhole must contain more than one pigeon. Formally, if n items are put into m containers and $n > m$, then at least one container must contain more than one item.

Given, $a_n = 5a_{n-1} - 6a_{n-2}$.

By comparing we get; $c_1 = 5$ and $c_2 = -6$

Then the characteristics eq^n is given as.

$$r^2 - c_1 r - c_2 = 0$$

$$\Rightarrow r^2 - 5r + 6 = 0$$

$$\Rightarrow r^2 - (3+2)r + 6 = 0$$

$$\Rightarrow r^2 - 3r - 2r + 6 = 0$$

$$\Rightarrow r(r-3) - 2(r-3) = 0$$

$$\Rightarrow (r-3)(r-2) = 0$$

$$\therefore r = 3 \text{ and } 2.$$

Thus, $a_n = \alpha_1 (3)^n + \alpha_2 (2)^n$

By given question:

for, $a_0 = 1$

$\Rightarrow 1 = \alpha_1 (3)^0 + \alpha_2 (2)^0$

$\Rightarrow 1 = \alpha_1 + \alpha_2$ — (i)

for $a_1 = 3$

$\Rightarrow 3 = 3\alpha_1 + 2\alpha_2$ — (ii)

$\Rightarrow$

Solving (i) and (ii).

$$3\alpha_1 + 2\alpha_2 = 3$$

$$\alpha_1 + \alpha_2 = 1 \quad \} \times 3$$

$$(-) \quad (-) \quad (-)$$

$$-\alpha_2 = 0$$

$$\text{or, } \alpha_2 = -0$$

Putting the value of α_2 in eq^n ①.

$$1 = \alpha_1 - 0$$

$$\Rightarrow \alpha_1 = 1$$

Substituting the value of α_1 and α_2 then,

$$a_n = 3^n + 0 \cdot 2^n$$

$$\therefore a_n = 3^n$$

So the solution to the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 1$, $a_1 = 3$ is:

$$a_n = 3^n \quad //$$

Section-B

4. Explain the principle of inclusion and exclusion. How many integers from 1 to 30 are multiples of 2 or 3.

→ Principle of inclusion and exclusion is a method in combinatorics to count the no. of elements in the union of multiple sets. It helps to avoid overcounting elements that belong to more than one set.

Formula: for two sets A and B.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

for three sets A, B and C:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| + |A \cap B \cap C|$$

Example: 30 students playing soccer, 20 students playing basketball & 10 students who play both. We have to find total no. of students who play at least one of these sports.

① Include: Add no. of soccer & basketball players.

$$30 + 20 = 50$$

② Exclude: Subtract the no. of students those who play both sports.

$$50 - 10 = 40.$$

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$\therefore$ By mathematical induction, the statement $1^3+2^3+3^3+\dots+n^3 = \frac{n^2(n+1)^2}{4}$ is true for all $n \geq 1$.

Section-B.

4. List any one example of tautology. Represent the following sentences into predicate logic:

a) Not all employees are loyal.

b) All students have good attitude are lovable.

→ A tautology is a statement that is true in every possible interpretation.

Example:

Either it will rain tomorrow or it will not rain tomorrow.

In logical terms, let P represent "it will rain tomorrow". The statement can be expressed as : $P \vee \neg P$.

This statement is always true because:

If P is true then $P \vee \neg P$ is true because P is true.

If P is false then $P \vee \neg P$ is true because $\neg P$ is true.

Therefore, "Either it will rain tomorrow, or it will not rain tomorrow" a tautology because it covers all possible scenarios & is always true.

a) Not all employees are loyal.

→ let $E(x)$: x is an employee.

$L(x)$: x is loyal.

$\therefore \neg \forall x \, (E(x) \rightarrow L(x))$

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b) All students having good attitude are lovable.

→ Predicates:

$S(x)$: x is a student

$A(x)$: x has a good attitude

$L(x)$: x is lovable.

$\therefore \forall x\,(S(x) \wedge A(x) \rightarrow L(x))$

5) Prove that "If the product of two integers a and b are even then either a is even or b is even" using the contradiction method.

→ Solution:

Step 1: Assuming that the product of ab is even but neither a nor is even.

Step 2: An integer is odd then it can be written as:

$a = 2k+1$ and $b = 2l+1$ where, k & l is an i

Step 3: Computing the product of a & b:

$$ab = (2k+1)(2l+1)$$
$$= 2k \cdot 2l + 2k \cdot 1 + 2l \cdot 1 + 1$$
$$= 4kl + 2k + 2l + 1$$
$$= 2(2kl + k + l) + 1$$

let's suppose $2kl + k + l$ as m then.

$$ab = 2m+1$$

Step 4: Since the expression $2m+1$ is the definition of odd number Therefore the product ab is odd.

Step 5: We assumed ab is even but our calculation shows ab is odd.

$\therefore$ Our assumption leads to a contradiction. so product of a & b a even then at least a or b must have to be even.

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6) Use Chinese Remainder Theorem to find the value of x such that

$$x = 0 \pmod 2$$
$$x = 2 \pmod 3$$
$$x = 3 \pmod 5$$

→ Solution:

$a_1 = 0 \qquad m_1 = 2$

$a_2 = 2 \qquad m_2 = 3$

$a_3 = 3 \qquad m_3 = 5$

Now, $M = m_1 \times m_2 \times m_3 = 2 \times 3 \times 5 = 30$

$M_1 = \frac{M}{m_1} = \frac{30}{2} = 15$, $M_2 = \frac{30}{3} = 10$, $M_3 = \frac{30}{5} = 6$.

Now,

$M_1 \cdot M_1^{-1} = 1 \pmod{m_1}$

$\Rightarrow 15 \cdot M_1^{-1} = 1 \pmod 2$

$\Rightarrow M_1^{-1} = 1$

$M_2 \cdot M_2^{-1} = 1 \pmod{m_2}$

$\Rightarrow 10 \cdot M_2^{-1} = 1 \pmod 3$

$\Rightarrow M_2^{-1} = 1$

$M_3 \cdot M_3^{-1} = 1 \pmod 5$

$\Rightarrow 6 \cdot M_3^{-1} = 1 \pmod 5$

$\Rightarrow M_3^{-1} = 1$

Now, $x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + a_3 M_3 M_3^{-1}) \bmod M$

$= (0 + 2 \times 10 \times 1 + 5 \times 6 \times 1) \bmod 30$

$\Rightarrow (20 + 30) \bmod 30$

$\Rightarrow 50 \bmod 30$

$\Rightarrow 20$

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Qn 7, 8 X X

Qn. 9 Give the premises "If it rains or strike holds then the exam will be cancelled. If it doesn't rain then it will be sunny day. The exam was not cancelled. Show that it is sunny day."

→ Solution:

let, R denotes it rains.
S denotes strike holds.
C denotes exam cancelled.
U denoted it is sunny day.

So from question: premise 1 : $(R \vee S) \rightarrow C$
premise 2 : $\neg R \rightarrow U$
premise 3 : $\neg C$

To prove: it is sunny day i.e. U.

Now, from premise 3 we know exam was not cancelled so, from premise 1

$(R \vee S) \rightarrow C$

contrapositive. $\neg C \rightarrow \neg(R \vee S)$

Since we know, $\neg C$ is true, it follows:

$\neg(R \vee S)$

By DeMorgan's law: $\neg R \wedge \neg S$

This means, it didn't rain and strike did not hol

By premise 2, $\neg R \rightarrow U$

Sine $\neg R$ is true by Modus Ponens: U

∴ It is a sunny day.

2. Find GCD

→ Solution:

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